OUTPUT:

Challenge 1

Downloading data for ['AAPL', 'MSFT', 'GOOG', 'AMZN', 'NVDA'] from 2020-01-01 to 2025-06-16...

<ipython-input-1-1936261411>:16: FutureWarning: YF.download() has changed argument auto_adjust default to True

data = yf.download(tickers, start=start_date, end=end_date, progress=False)

Error downloading data: 'Adj Close'

Falling back to dummy data for demonstration.

Using dummy data for demonstration.

Sample of dummy daily returns:

	AAPL	MSFT	GOOG	AMZN	NVDA
2020-01-01	0.004967	-0.001383	0.006477	0.015230	-0.002342
2020-01-02	-0.002341	0.015792	0.007674	-0.004695	0.005426
2020-01-03	-0.004634	-0.004657	0.002420	-0.019133	-0.017249
2020-01-06	-0.005623	-0.010128	0.003142	-0.009080	-0.014123
2020-01-07	0.014656	-0.002258	0.000675	-0.014247	-0.005444

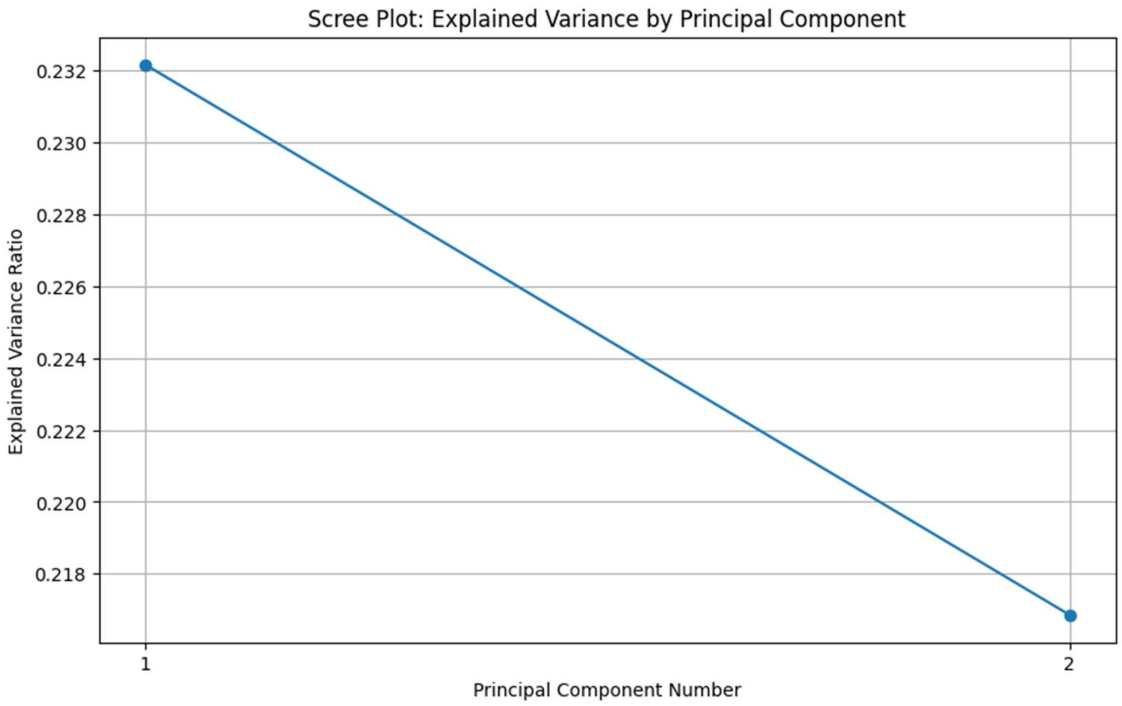
Original Data Shape: (250, 5)

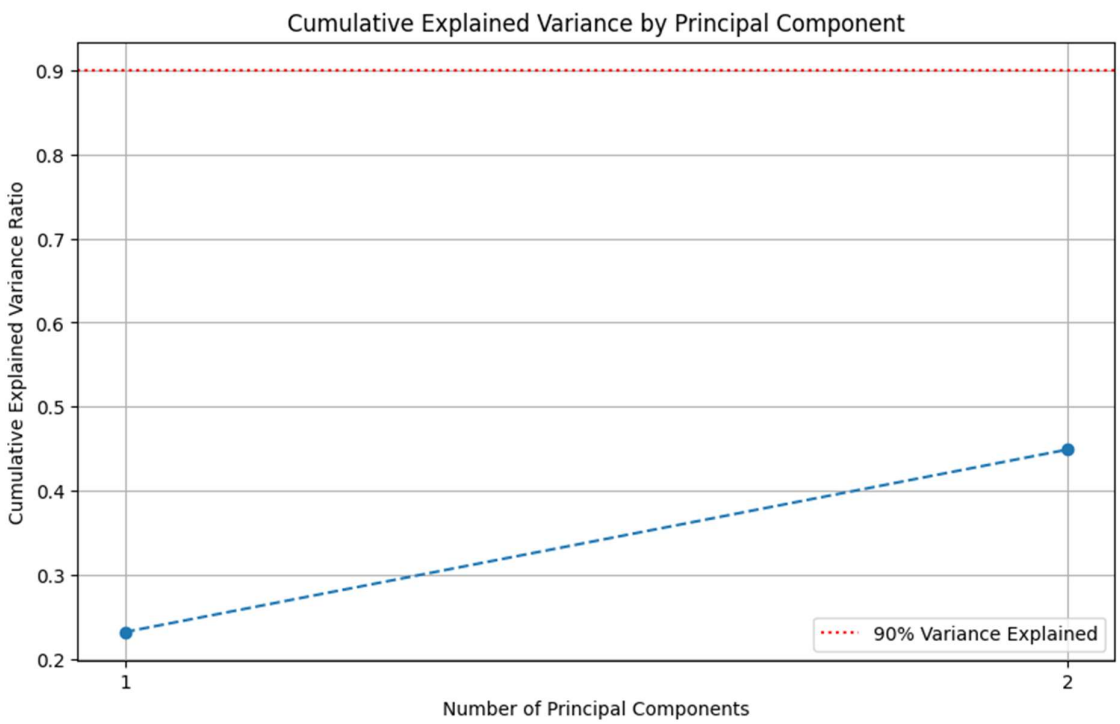
Sample of Scaled Returns:

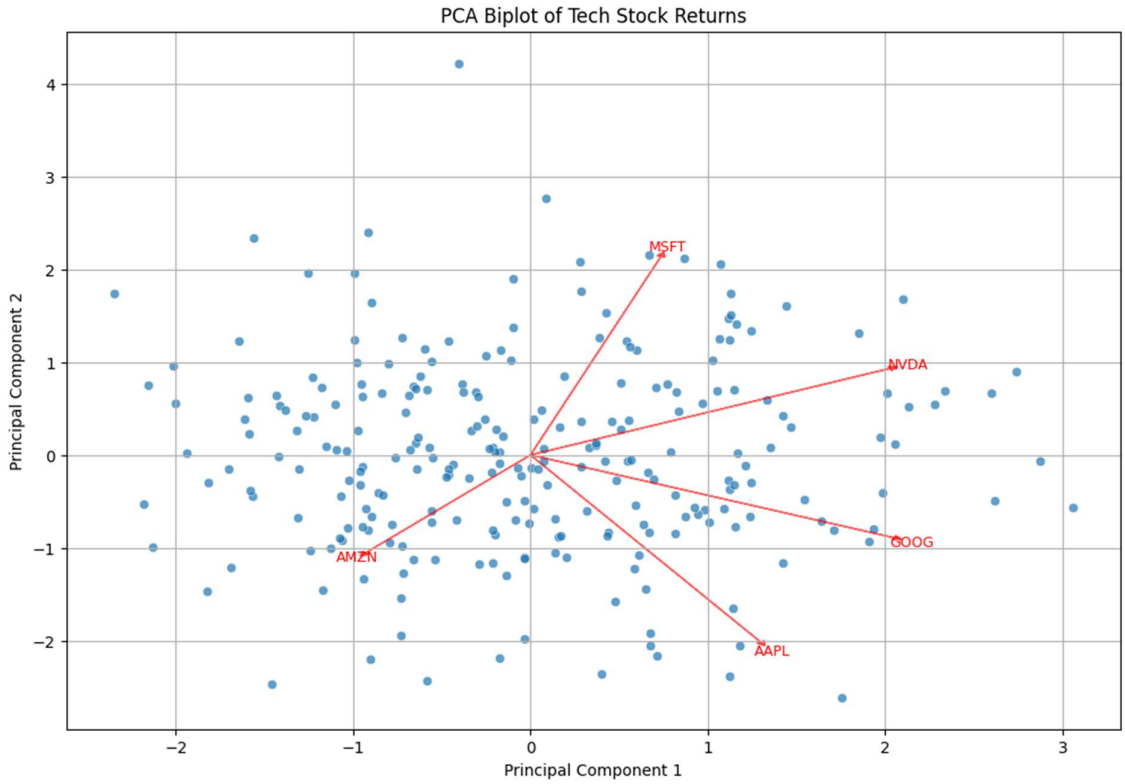
	AAPL	MSFT	GOOG	AMZN	NVDA
2020-01-01	0.511921	-0.093822	0.667288	1.478569	-0.343482
2020-01-02	-0.269829	1.568362	0.791827	-0.572084	0.411382
2020-01-03	-0.515078	-0.410745	0.245321	-2.058026	-1.792310
2020-01-06	-0.620833	-0.940233	0.320499	-1.023432	-1.488490
2020-01-07	1.548335	-0.178517	0.063904	-1.555236	-0.644984

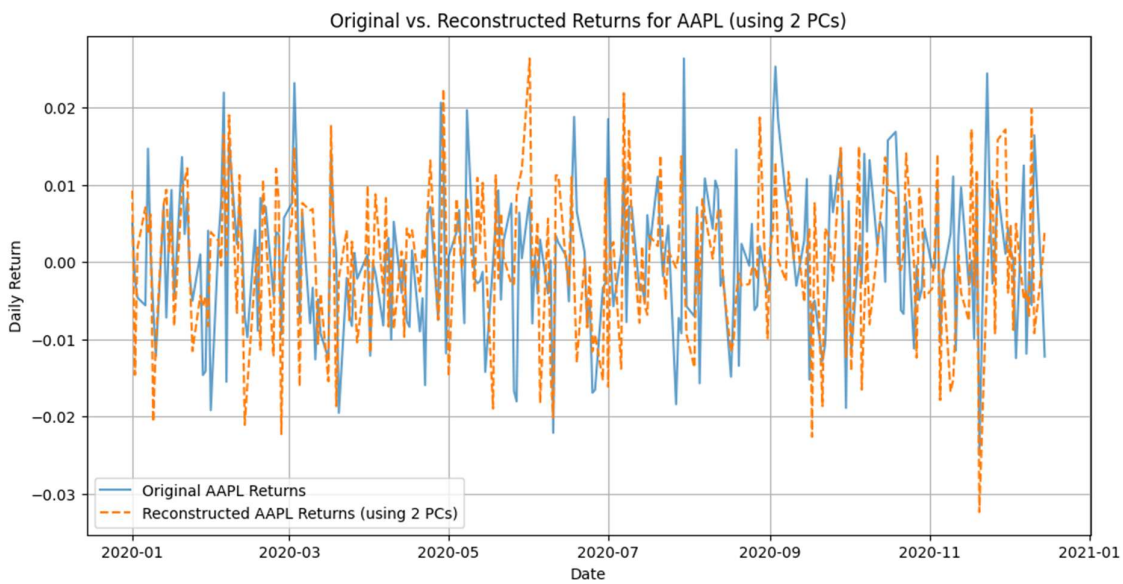
Extracted Principal Components (First 2 components):

```
                PC_1      PC_2
2020-01-01 -0.031339 -1.097103
2020-01-02  1.123514  1.237196
2020-01-03 -0.647951  0.130088
2020-01-06 -0.860641 -0.401091
2020-01-07  0.635569 -0.745915
New Data Shape (after PCA with 2 components): (250, 2)

Explained Variance Ratio for each Principal Component:
  PC_1: 0.2322 (23.22%)
  PC_2: 0.2169 (21.69%)
  Cumulative explained variance by 2 components: 0.4490

Principal Component Loadings (how original features contribute to PCs):
          AAPL      MSFT     GOOG      AMZN      NVDA
PC_1  0.384230  0.218025  0.60742 -0.275103  0.600159
PC_2 -0.596389  0.633327 -0.26401 -0.312089  0.275891
```


Scree Plot: Explained Variance by Principal Component
Explained Variance Ratio
0.232
0.230
0.228
0.226
0.224
0.222
0.220
0.218
1
2
Principal Component Number


Cumulative Explained Variance by Principal Component
Cumulative Explained Variance Ratio
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
90% Variance Explained
1
2
Number of Principal Components





Mean Squared Error of Reconstruction for each feature:

9.660822565600153e-05

Average MSE across all features: 0.000097

Challenge 3

Downloading data for ['^GSPC', '^IXIC', '^DJI', '^RUT'] from 2020-01-01 to 2025-06-16...

<ipython-input-3-1663625532>:20: FutureWarning: YF.download() has changed argument auto_adjust default to True

data_multi = yf.download(tickers_multi, start=start_date_multi, end=end_date_multi, progress=False)

Error downloading data: 'Adj Close'

Falling back to dummy data for demonstration.

Using dummy data with multicollinearity for demonstration.

Sample of dummy index prices:

	^GSPC	^IXIC	^DJI	^RUT
2020-01-01	999.624693	1199.782640	899.972003	800.464232
2020-01-02	1000.282871	1199.743293	899.685211	800.261149
2020-01-03	1000.905941	1199.804937	899.288441	799.992357
2020-01-06	1000.103483	1199.371925	899.163462	800.056759
2020-01-07	999.369412	1198.556892	899.212487	800.099779

--- OLS Regression Results (Potentially Affected by Multicollinearity) ---

OLS Regression Results

==========

Dep. Variable: Target_RUT_NextDay R-squared: 0.013

Model: OLS Adj. R-squared: 0.001

Method: Least Squares F-statistic: 1.105

Date: Tue, 17 Jun 2025 Prob (F-statistic): 0.348

Time: 14:37:40 Log-Likelihood: 1609.4

No. Observations: 248 AIC: -3211.

Df Residuals: 244 BIC: -3197.

Df Model: 3

```
Covariance Type:          nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const     -4.377e-05   2.37e-05     -1.844      0.066   -9.05e-05    2.98e-06
^GSPC        -0.0540      0.046     -1.168      0.244      -0.145       0.037
^IXIC         0.0578      0.049      1.185      0.237      -0.038       0.154
^DJI         -0.0516      0.052     -0.985      0.326      -0.155       0.052
==============================================================================
Omnibus:                        2.285   Durbin-Watson:                   2.072
Prob(Omnibus):                  0.319   Jarque-Bera (JB):                2.275
Skew:                           0.231   Prob(JB):                        0.321
Kurtosis:                       2.921   Cond. No.                     2.27e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.27e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

--- Variance Inflation Factors (VIF) Scores ---
  feature       VIF
0   ^GSPC  1.007030
1   ^IXIC  1.009335
2    ^DJI  1.006176
```

--- Ridge Regression Results (Handling Multicollinearity) ---

Ridge Regression with alpha = 0.1

Coefficients (on scaled data):

  ^GSPC: -0.0000

  ^IXIC: 0.0000

  ^DJI: -0.0000

Intercept (on scaled data): -0.0001

RMSE on Test Set: 0.0003

Ridge Regression with alpha = 1.0

Coefficients (on scaled data):

  ^GSPC: -0.0000

  ^IXIC: 0.0000

  ^DJI: -0.0000

Intercept (on scaled data): -0.0001

RMSE on Test Set: 0.0003

Ridge Regression with alpha = 10.0

Coefficients (on scaled data):

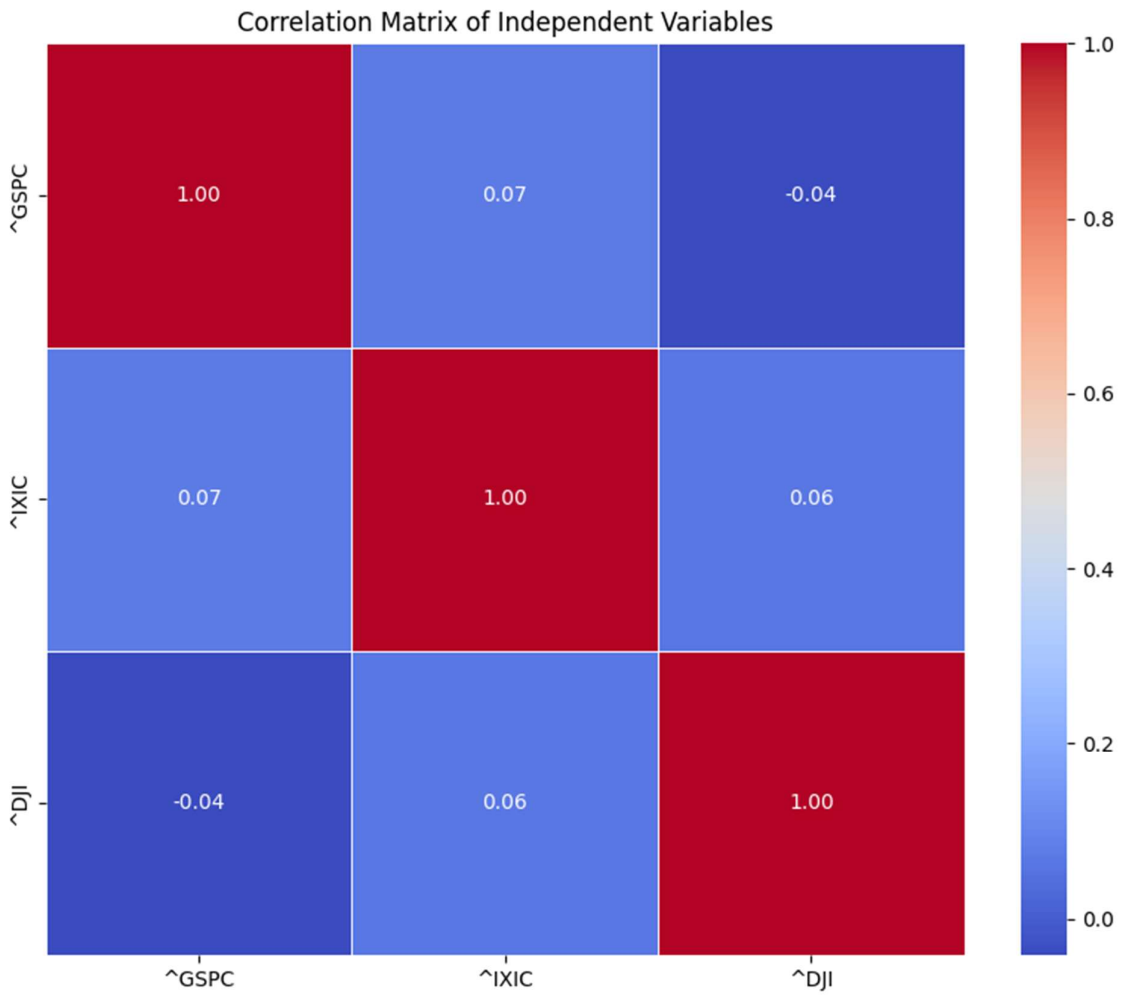
  ^GSPC: -0.0000

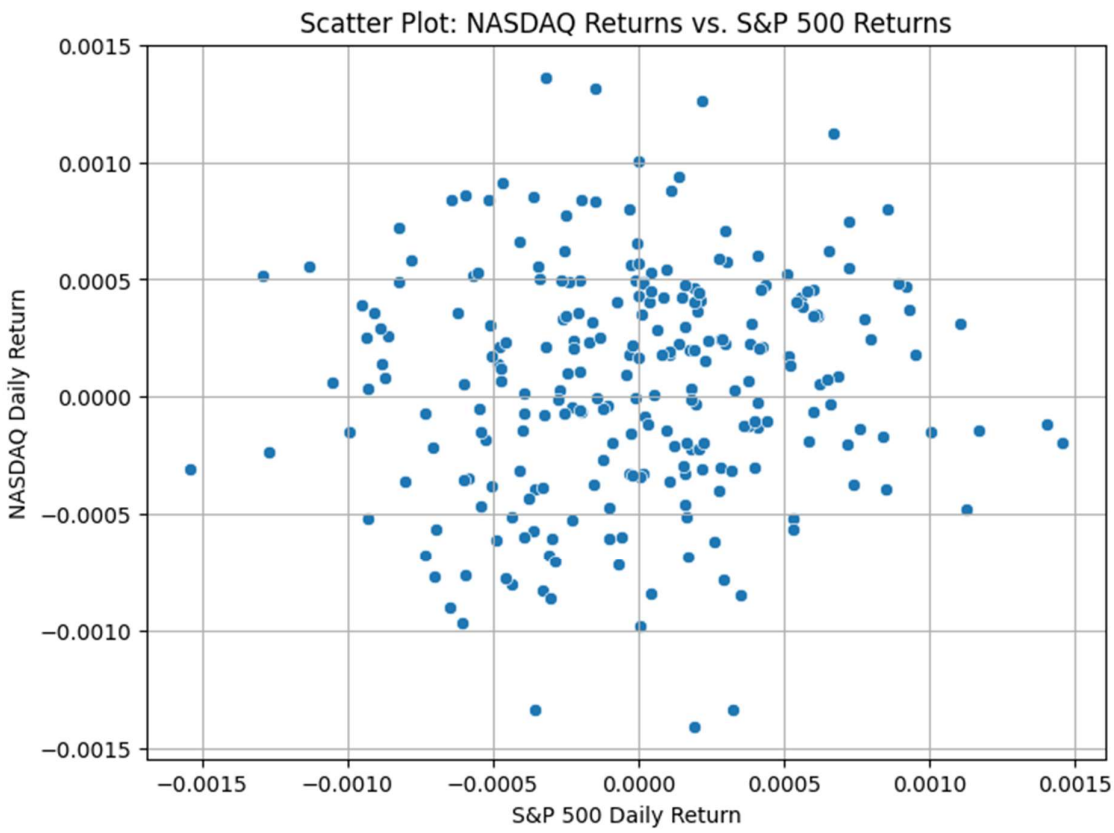
  ^IXIC: 0.0000

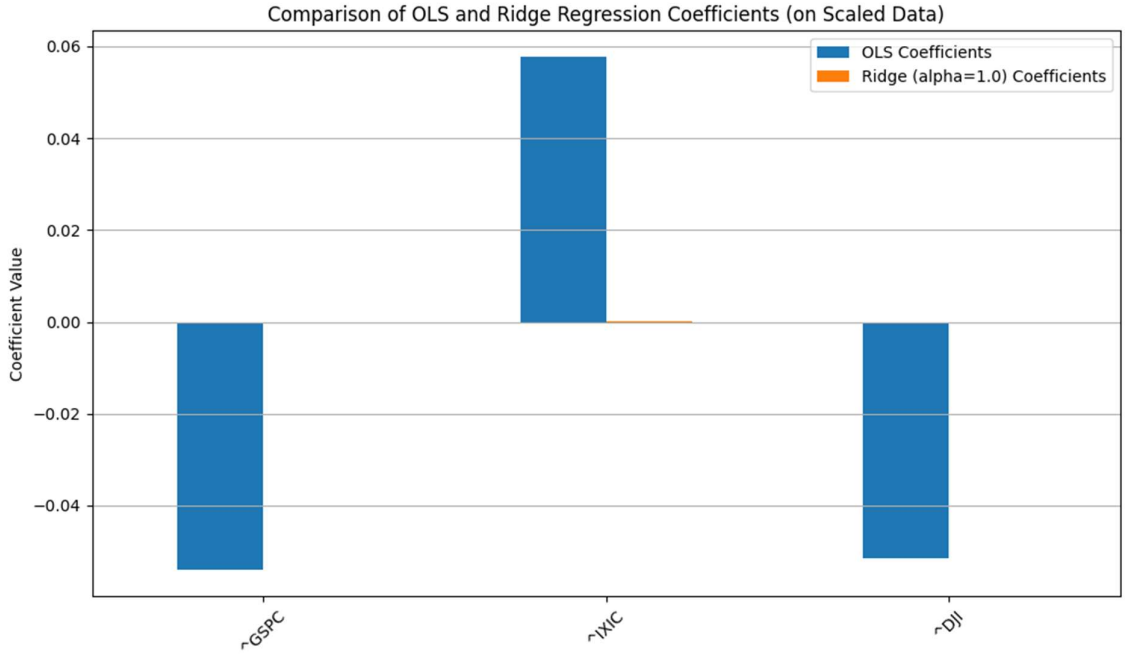
  ^DJI: -0.0000

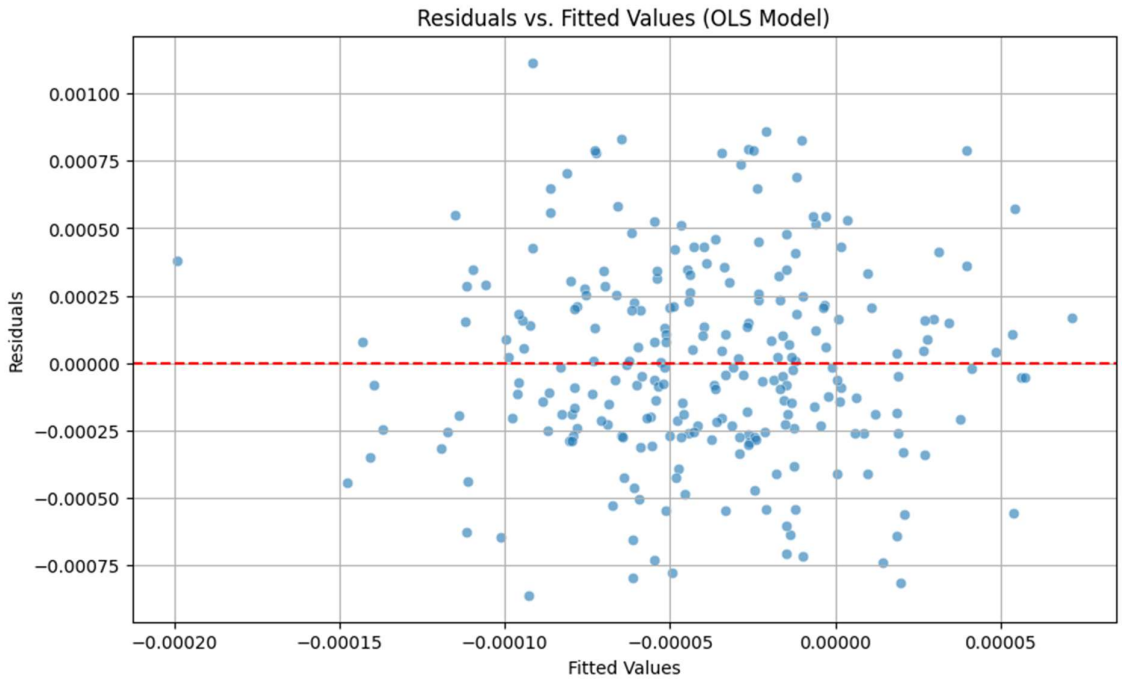
Intercept (on scaled data): -0.0001

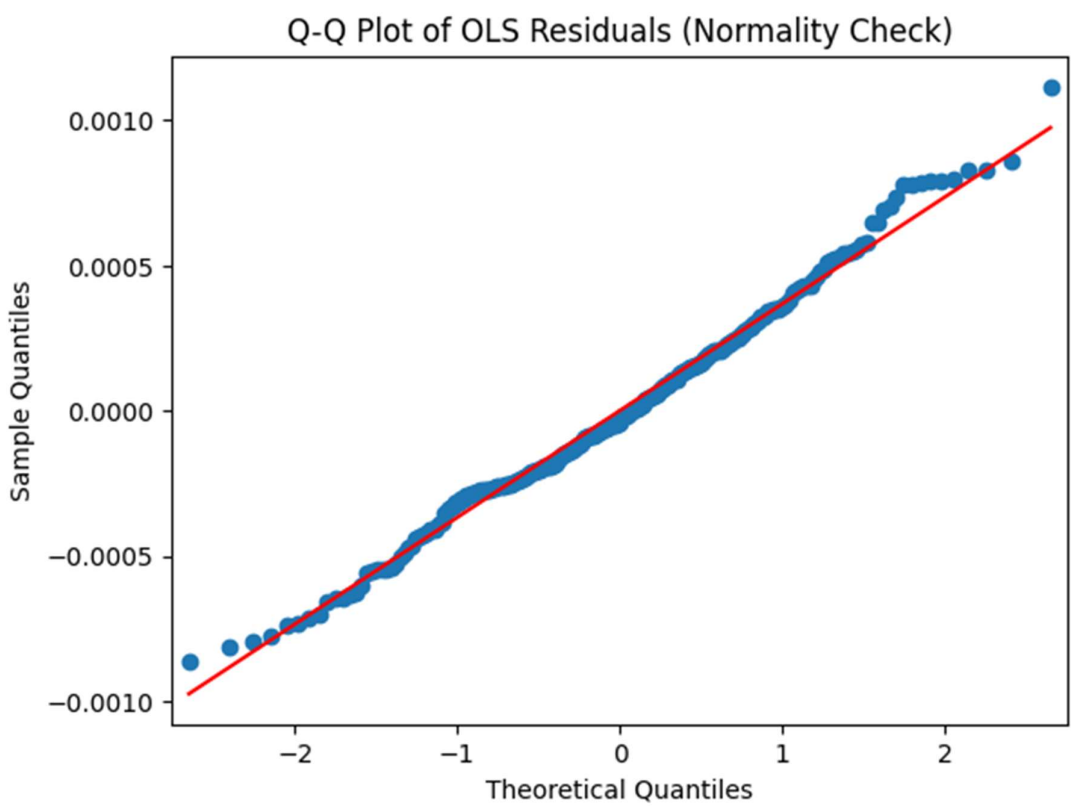
RMSE on Test Set: 0.0003


Correlation Matrix of Independent Variables
^GSPC
^IXIC
^DJI
1.00
0.07
-0.04
0.07
1.00
0.06
-0.04
0.06
1.00
^GSPC
^IXIC
^DJI
1.0
0.8
0.6
0.4
0.2
0.0


Scatter Plot: NASDAQ Returns vs. S&P 500 Returns
NASDAQ Daily Return
S&P 500 Daily Return
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015


Comparison of OLS and Ridge Regression Coefficients (on Scaled Data)
OLS Coefficients
Ridge (alpha=1.0) Coefficients
Coefficient Value
0.06
0.04
0.02
0.00
−0.02
−0.04
^GSPC
^IXIC
^DJI





RMSE on Test Set (OLS Model): 0.0003

RMSE on Test Set (Ridge Model with alpha=1.0): 0.0003

Conclusion: Ridge regression improved out-of-sample performance (lower RMSE).